

Title: Project Registration & Progress Review

Department: Computer Engineering	Academic Year: 2026-27
Semester: 5	Group No.: TY CS H-13
Project Title: Medlink: An Adaptive Offline-First Telemedicine Platform with Intelligent Network-Aware Consultation Switching	
Project Area: Telemedicine Systems, Mobile Application Development, Offline-First Computing, Healthcare Informatics.	

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Project approved / Not approved

Guide

Project Coordinator

Head of Department

Project Synopsis:**1. Abstract**

Access to quality healthcare in rural and remote regions remains a significant challenge due to unreliable internet connectivity, which limits the effectiveness of conventional telemedicine platforms. Existing solutions primarily rely on stable network conditions for real-time video consultations, often resulting in interrupted or failed consultations in low-bandwidth environments. **Medlink** proposes an adaptive, offline-first telemedicine platform that ensures uninterrupted healthcare delivery by intelligently adjusting consultation modes based on real-time network conditions. The platform continuously monitors bandwidth and latency to seamlessly transition between video consultation, audio consultation, asynchronous voice and image-based communication, and complete offline data capture with automatic synchronization upon network restoration. Patient health records, appointment requests, prescriptions, and consultation history are securely stored locally using an offline-first architecture and synchronized with a centralized cloud database when connectivity becomes available. A doctor dashboard provides real-time visibility into each patient's consultation mode, enabling efficient and uninterrupted communication.

2. Introduction

Rural and remote regions often face challenges in accessing quality healthcare due to unreliable internet connectivity and limited medical infrastructure. Existing telemedicine platforms depend on stable network connections, limiting their effectiveness in low-bandwidth environments. **MedLink** addresses this problem by combining adaptive consultation switching, offline-first data synchronization, and intelligent network-aware communication to provide a reliable and accessible telemedicine solution.

3. Problem Statement

Most telemedicine platforms assume stable internet connectivity and rely on continuous network access for real-time consultations. They also provide limited support for offline data access, causing interrupted consultations and reduced healthcare accessibility in rural areas. There is a need for a reliable, adaptive telemedicine solution that supports seamless consultation switching and offline-first healthcare services.

4. Objectives

1. Develop an adaptive telemedicine platform that enables uninterrupted healthcare consultations by automatically switching between video, audio, asynchronous messaging, and offline modes based on network conditions.
2. Build an offline-first healthcare system that securely stores patient records, appointments, prescriptions, and consultation history locally with automatic cloud synchronization upon connectivity restoration.
3. Design a network-aware doctor dashboard that provides real-time consultation status and supports efficient patient management across multiple communication modes.
4. Integrate intelligent bandwidth monitoring and synchronization mechanisms to ensure reliable healthcare delivery, seamless data consistency, and improved accessibility for rural and remote communities.

5. Literature Survey

Existing telemedicine platforms primarily rely on:

- Stable Internet Connectivity
- Real-Time Video Consultations
- Cloud-Based Patient Record Storage
- Manual Appointment Management
- Conventional Communication Methods

Although these approaches improve access to remote healthcare services, they depend on continuous network availability and provide limited offline functionality. They also lack adaptive consultation switching and resilient offline-first data synchronization for low-connectivity rural environments.

Sr no.	Name of paper	Year of publication	Name of authors	Journal/Conference Name	Methodology	Findings	Research Gap
1	Reimagining India's National Telemedicine Service to Improve Access to Care	2024	Ashish Jha et al.	The Lancet Regional Health – Southeast Asia	Evaluated India's eSanjeevani telemedicine platform, analyzing its workflow, tele-referral system, and implementation challenges.	eSanjeevani has enabled millions of consultations but faces issues such as outdated technological support, poor referral workflows, and inadequate digital infrastructure	Does not provide adaptive consultation switching or offline-first functionality for unreliable rural networks..
2	Improving the Effectiveness of Telemedicine in Rural Communities: Reflecting on Client and Provider Experience of eSanjeevani Jharkhand	2023	Aditya Naskar, Shilpa Bhatte, Neha Verma, Amit Parab, et al.	Population Medicine	Studied patient and healthcare provider experiences using eSanjeevani in rural Jharkhand.	Telemedicine improved healthcare accessibility, but connectivity issues, digital literacy, and operational challenges reduced service effectiveness.	Lacks offline consultation support and intelligent adaptation to poor network conditions.
3	A Cross-Sectional Study on Utilization and Barriers of eSanjeevani Telemedicine Services in Rural Areas of South India	2024	Kavyashree C, Dr. Nagaveni S. J.	Bharati Vidyapeeth Medical Journal	Conducted a rural survey to identify utilization patterns and barriers affecting telemedicine adoption.	rural survey to identify utilization patterns and barriers affecting telemedicine adoption. Identified poor internet connectivity, limited awareness, and technical difficulties as major barriers to telemedicine usage.	Existing systems do not support automatic consultation mode switching or offline record synchronization.

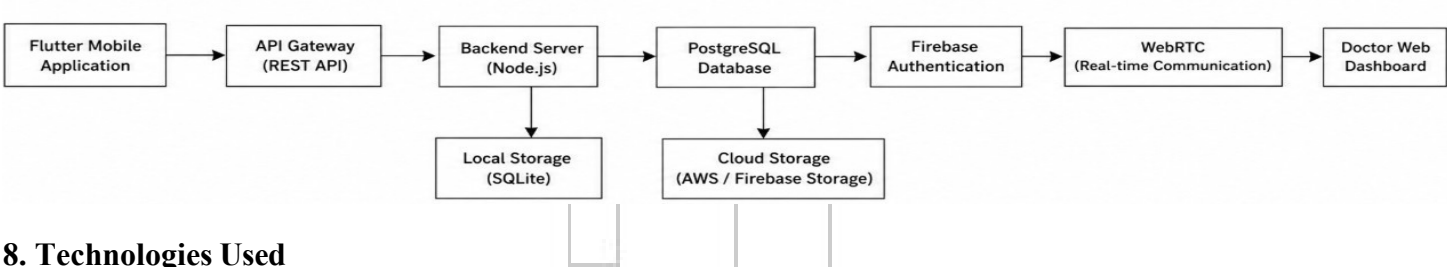
4	Challenges, Barriers, and Facilitators in Telemedicine Implementation in India: A Scoping Review	2024	Multiple Authors	PMC (Scoping Review)	Reviewed 26 studies published between 2019 and 2024 on telemedicine adoption in India.	Highlighted unreliable internet, digital literacy, infrastructure limitations, and lack of standardized telemedicine systems as major implementation challenges.	Existing solutions lack resilient offline-first architecture and adaptive communication mechanisms for rural healthcare.
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6. Proposed System

MedLink consists of two integrated components:

- Adaptive Consultation Management:** The platform continuously monitors network conditions and automatically switches between video, audio, asynchronous voice/image messaging, and offline consultation modes to ensure uninterrupted healthcare services.
- Offline-First Healthcare Synchronization:** Patient records, appointments, prescriptions, and consultation history are securely stored locally and automatically synchronized with the cloud when connectivity is restored, enabling reliable and continuous access to healthcare information.

7. System Architecture



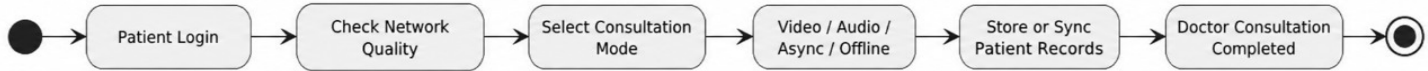
8. Technologies Used

Layer	Technology
Frontend	Flutter
Backend	Node.js
Database	PostgreSQL
Authentication	Firebase
Local Storage	SQLite
State Management	Provider / Riverpod
Real-Time Communication	Web RTC
API	REST API

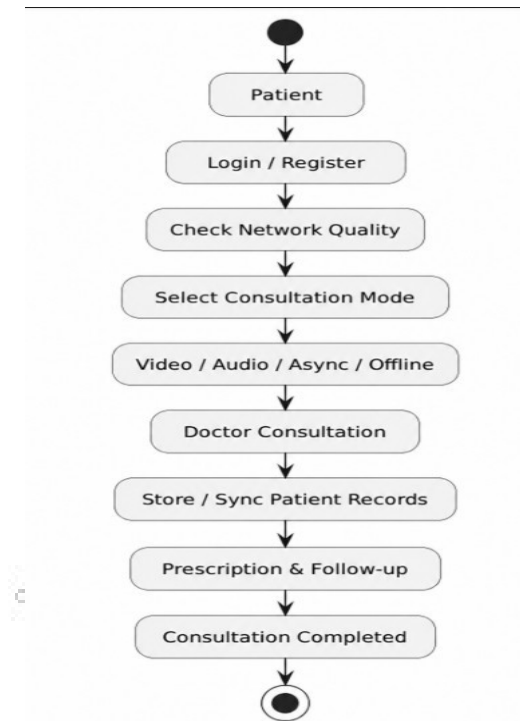
9. Workflow

Map Creation:

Patient Consultation Workflow:



Navigation



10. Algorithms

- **Adaptive Bandwidth Monitoring** – Continuously evaluates network quality to determine the optimal consultation mode.
- **Consultation Mode Switching** – Automatically transitions between video, audio, asynchronous messaging, and offline modes based on connectivity.
- **Offline Data Synchronization** – Synchronizes locally stored patient records with the central database when connectivity is restored.
- **Conflict Resolution** – Ensures data consistency by resolving simultaneous updates made offline and online.
- **Authentication & Authorization** – Provides secure access control for patients and healthcare professionals.

11. Expected Outcomes

- Adaptive telemedicine consultations based on real-time network conditions.
- Uninterrupted healthcare services in low-bandwidth and offline environments.
- Secure offline storage and automatic synchronization of patient records.
- Reliable video, audio, and asynchronous consultation support.
- Improved healthcare accessibility for rural and remote communities.
- Efficient doctor dashboard for patient monitoring and consultation management.
- Enhanced continuity, reliability, and accessibility of telemedicine services.

12. Sustainable Development Goals(SDGs) Addressed

Goal	How this project contributes
SDG 3 — Good Health and Well-being	Delivers encrypted video consultations, digital prescription issuance, and appointment management — extending access to verified clinical care beyond the boundary of physical proximity.
SDG 10 — Reduced Inequalities	Eliminates the geographic and economic barrier to specialist consultation. A patient in a remote or underserved area can access a licensed physician through the same platform as an urban user, with no difference in care quality.
SDG 9 — Industry, Innovation & Infrastructure	Builds a role-scoped digital health infrastructure (Doctor Portal, Patient App, Pharmacy Portal) on a shared REST + WebSocket + WebRTC backend — demonstrating that clinical-grade telemedicine can be built on open, modern web infrastructure.
SDG 17 — Partnerships for the Goals	Creates a multi-stakeholder collaboration layer connecting patients, physicians, and pharmacists under a single governed platform — enabling end-to-end care coordination that no single actor could deliver independently.

Group No.	TY CS H13		
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1	Group No.:	Date:																																																									
Progress Review Report Project Progress Checklist <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Parameter</th> <th>Parameter status Description</th> <th>Status</th> <th>Remark</th> </tr> </thead> <tbody> <tr> <td>Problem Statement</td> <td>Clearly defined</td> <td>☐ Complete ☐ Partial</td> <td></td> </tr> <tr> <td>Literature Review</td> <td>Relevant papers studied</td> <td>☐ Complete ☐ Partial</td> <td></td> </tr> <tr> <td>Research Gap</td> <td>Identified</td> <td>☐ Complete ☐ Partial</td> <td></td> </tr> <tr> <td>Objectives</td> <td>Measurable objectives</td> <td>☐ Complete ☐ Partial</td> <td></td> </tr> <tr> <td>Requirements</td> <td>SRS/requirements prepared</td> <td>☐ Complete ☐ Partial</td> <td></td> </tr> <tr> <td>System Design</td> <td>Architecture/design completed</td> <td>☐ Complete ☐ Partial</td> <td></td> </tr> <tr> <td>Dataset</td> <td>Collected/prepared</td> <td>☐ Complete ☐ Partial ☐ N/A</td> <td></td> </tr> <tr> <td>Methodology</td> <td>Finalized</td> <td>☐ Complete ☐ Partial</td> <td></td> </tr> </tbody> </table> Quality Check – Mid-Semester <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Quality Factor</th> <th>Review Questions</th> <th>Status</th> </tr> </thead> <tbody> <tr> <td>Environment</td> <td>Has resource/environmental impact been considered?</td> <td>☐ Yes ☐ No</td> </tr> <tr> <td>Sustainability</td> <td>Is the system maintainable and scalable?</td> <td>☐ Yes ☐ No</td> </tr> <tr> <td>Safety</td> <td>Have potential safety risks been identified?</td> <td>☐ Yes ☐ No</td> </tr> <tr> <td>Security</td> <td>Have security threats and controls been identified?</td> <td>☐ Yes ☐ No</td> </tr> <tr> <td>Ethics</td> <td>Are privacy, consent, fairness and IP considered?</td> <td>☐ Yes ☐ No</td> </tr> <tr> <td>Cost</td> <td>Has feasibility/cost been estimated?</td> <td>☐ Yes ☐ No</td> </tr> </tbody> </table> Guide Comments: Signature of Guide:			Parameter	Parameter status Description	Status	Remark	Problem Statement	Clearly defined	☐ Complete ☐ Partial		Literature Review	Relevant papers studied	☐ Complete ☐ Partial		Research Gap	Identified	☐ Complete ☐ Partial		Objectives	Measurable objectives	☐ Complete ☐ Partial		Requirements	SRS/requirements prepared	☐ Complete ☐ Partial		System Design	Architecture/design completed	☐ Complete ☐ Partial		Dataset	Collected/prepared	☐ Complete ☐ Partial ☐ N/A		Methodology	Finalized	☐ Complete ☐ Partial		Quality Factor	Review Questions	Status	Environment	Has resource/environmental impact been considered?	☐ Yes ☐ No	Sustainability	Is the system maintainable and scalable?	☐ Yes ☐ No	Safety	Have potential safety risks been identified?	☐ Yes ☐ No	Security	Have security threats and controls been identified?	☐ Yes ☐ No	Ethics	Are privacy, consent, fairness and IP considered?	☐ Yes ☐ No	Cost	Has feasibility/cost been estimated?	☐ Yes ☐ No
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Review No.: 2 **Group No.:** **Date:**

Progress Review Report

Final Progress Review Check list

Component	Final project Status Completed/Not Completed	Remark
Implementation	☐ Yes ☐ No	
Database/API	☐ Yes ☐ No ☐ N/A	
Testing	☐ Yes ☐ No	
Performance evaluation	☐ Yes ☐ No	
Security testing	☐ Yes ☐ No	
Safety testing	☐ Yes ☐ No ☐ N/A	
Documentation	☐ Yes ☐ No	
User evaluation	☐ Yes ☐ No ☐ N/A	
Final working prototype	☐ Yes ☐ No	

Final Quality Check

Quality Factor	Evidence Expected	Submitted
Environment	Resource/environment impact analysis	☐ Yes ☐ No
Sustainability	Maintenance, scalability, upgrade plan	☐ Yes ☐ No
Safety	Safety test case result	☐ Yes ☐ No
Security	Security test case results	☐ Yes ☐ No
Ethics	Privacy, consent documents	☐ Yes ☐ No
Cost	Actual project cost bills/invoices	☐ Yes ☐ No

Guide Comments:

Signature of Guide: